Reflections on the Crofton formula

Work in progress

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Abstract: The Crofton formula (the length of an arc can be estimated by the number of hits by a random line) is extremely important in modern-day microscopy. Exemplifying Stigler's (1980) law of eponymy (itself actually due to Robert Merton[1]), the formula is more properly described as the Cauchy-Crofton formula (announced in *Comptes Rendus* by Cauchy (1832)), and originates from the famous thought experiment on Buffon's baguettes. Together with the associated Poincaré kinematic formula, Crofton's formula can be proved by a simple intuitive visual argument in the case of polygonal curves. But the Crofton and Poincaré results actually hold for the far more general case of curves of finite total variation, in which context they require rather more technical proofs. I will report on progress I have made in deriving intuitive visual arguments for the general case.

[1] Wikipedia:
"Everything of importance has been said before by somebody who did not discover it", attributed to Alfred North Whitehead.

History (1)

Figure 1: Comte de Buffon

Georges Louis Leclerc (Comte de Buffon, 1707-1788): from a family of minor civil servants, inherited a fortune from godfather, moved to Paris in 1732, researched mathematics and then geology and botany, made a (hereditary) count in 1772, both empiricist ("Anglo-Saxon") and rationalist ("Gallic"). Seeking a motivation for probability other than gambling, Buffon introduced his famous *baguette example* (now "Buffon's needle": see Comte de Buffon, 1829; translation, Hey *et al.*, 2010).

History (2)

Figure 2: Cauchy

Augustin-Louis Cauchy (1789-1857): French mathematician, engineer and physicist. He "is one of the 72 scientists whose name is inscribed on the first floor of the Eiffel Tower". Cauchy (1841; also 1832) used Buffon's ideas to show how to recover the length of a polygonal curve using *projections of the component segments on a random test line*.

History (3)

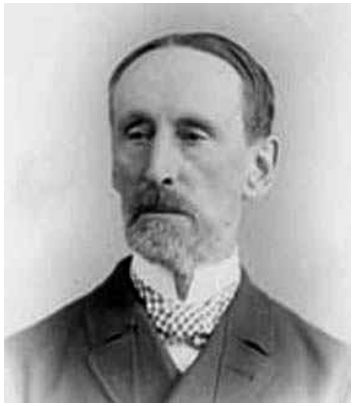

Figure 3: Crofton

Morgan William Crofton (1826–1915): Anglo-Irish, Anglican Protestant upbringing but converted to Roman Catholism (connected with John Henry Newman), Professor of Mathematics at the Royal Military Academy at Woolwich, famous for his long article on probability in ninth edition of Encyclopaedia Britannica (Crofton, 1902; also Crofton, 1868), including *recovery of length of perimeter of a convex polygon using intersections with a random line*.

Representative modern problems

The Cauchy-Crofton formula is fruitful in both applied and pure mathematics.

- (a) *Stereology:* impose random lattices on image sequence from microscope, count mean number of intersections with filaments of interest, deduce mean length of such a filament;
- (b) *Stochastic geometry:* consider mean number of intersections of curve with a *standard Poisson line process*, deduce length of curve;
- (c) *Differential geometry:* Chern's kinematic formula (for a very readable account, see Calegari, 2020!).

Crofton's formula: a visual proof

A standard Poisson line process Π in the plane is produced by a unit intensity Poisson point process on $\mathbb{R} \times [0, \pi)$ using intensity measure $\frac{1}{2}\mathrm{d}r\,\mathrm{d}\theta$: here θ is slope and r is the signed distance from origin of the line corresponding to (r, θ). The number of intersections of Π with a unit segment has unit mean (*simple calculation*).

Let $f{:}[0, L] \to \mathbb{R}^2$ be a polygonal curve, and consider its mean number of intersections with Π. Divide trajectory of f into variable length segments s:

$$\begin{aligned} \mathbb{E}\left[\#\left\{\mathrm{Image}(f) \cap \Pi\right\}\right] &= \sum_s \mathbb{E}\left[\#\left\{s \cap \Pi\right\}\right] \\ &= \sum_s \mathrm{length}(s) \end{aligned}$$

hence Crofton formula:
mean number of intersections with Π *equals length of curve*.

The above proof generalizes to yield *Poincaré's kinematic formula*:

Replace Π by a standard Poisson process of random polygonal fibres.

Crofton and Poincaré results generalize to piece-wise C^1 curves by (slightly untidy) approximation.

Can we generalize further?

Maximal generalization for Crofton formula

The components of Crofton's formula still make sense if $f{:}[0,L]\to\mathbb{R}^2$ is a continuous curve of finite length (Lipschitz(1)). How to prove this?

Options:

Ⓐ Use the *co-area formula* (Federer, 1969): Consider Lipschitz $p : M \to N$ between (eg) compact C^1 Riemannian manifolds of dimensions $m \geq n$. Then (adjusting exposition of Baddeley, 1983),

- for almost all $x \in N$, the fibre $p^{-1}(x)$ is the union of $(m{-}n)$-dimensional C^1 submanifolds (up to a null-set for $(m{-}n)$-Hausdorff measure);
- there is a measurable $J : M \to \mathbb{R}$ such that

$$\int_M f(z)J(z)\mathcal{H}^m(\mathrm{d}z) \quad = \quad \int_N \left(\int_{p^{-1}(x)} f(z)\mathcal{H}^{m-n}(\mathrm{d}z) \right) \mathcal{H}^n(\mathrm{d}x)\,.$$

Baddeley (1983) describes how to use the co-area formula to derive not only Crofton formula for Lipschitz curves but also elegant generalizations which are very useful in stereology.

Figure 4: Sledge hammer

However, using the co-area formula to prove the Crofton formula itself is rather like using a sledgehammer to crack a nut.

Ⓑ Ayari & Dubuc (1997) describe how to use the elegant exposition of total variation by Cesari (1958), together with a simple observation by Cauchy (1841; also 1832), which reduces the Crofton formula to the following *one-dimensional* version:

Banach's (1925) representation of total variation

For continuous $f{:}[0,L]{\to}\mathbb{R}$,

$$\text{Variation}\,(f) \quad = \quad \int_{\mathbb{R}} \#\,\{x \in [0,L] \;:\; f(x){=}y\}\ \mathrm{Leb}(\mathrm{d}y)\,, \qquad (1)$$

in the extended sense
(equality also holds when both sides of the equation are infinite).
The right-hand side counts the mean number of intersections of the graph of f with a standard Poisson line process of *horizontal* lines (indexed by a unit intensity Poisson point process on the y-axis).

Seeking a visual proof (1)

Objective: obtain a proof of Equation 1 that is as visual as possible.

Initial observation

1. **Infinite case:** infinite variation is straightforward (polygonal approximation lengths diverge).
2. **Symmetries:** If variation is finite, employ symmetry by
   (i) vertical shifts,
   (ii) non-linear change of x-scale
   to reduce to case in which $f(0) = 0$ and f has weak derivative $\dot{f}$ with $|\dot{f}| = 1$ almost everywhere on $[0, L]$
   (note L now equals total variation of f).

Seeking a visual proof (2)

3. **Un-folds:** an unfold based at $u<v\in[0,L]$
   i. reflects $f|_{[u,L]}$ in the horizontal line $y=f(u)$,
   ii. then reflects the resulting $\tilde{f}|_{[v,L]}$ in the horizontal line $y=f(v)$.
   (Equivalently, vertical reflection for $f|_{[u,v]}$ and vertical shift for $f|_{[v,L]}$.)

We only use unfolds for which $f(u)>f(v)$, and we say that the unfold has *size* $2(f(u)-f(v))$, which is the length of the vertical shift for $f|_{[v,L]}$.

The *maximal* unfold chooses $u<v$ to be the left-most pair realizing $\sup_{u\le v}(f(u)-f(v))$.

Seeking a visual proof (3)

Original path with $u<v$ chosen to yield the maximal unfold:

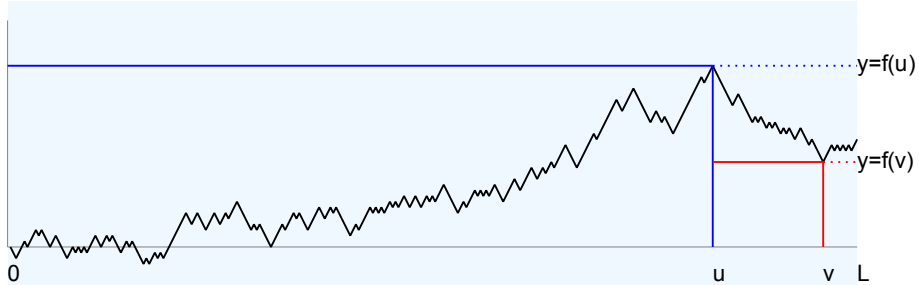

Figure 5: Maximal unfold based at $u<v$

Size of unfold is $2(f(u)-f(v))$.

Resulting unfolded path f_1:

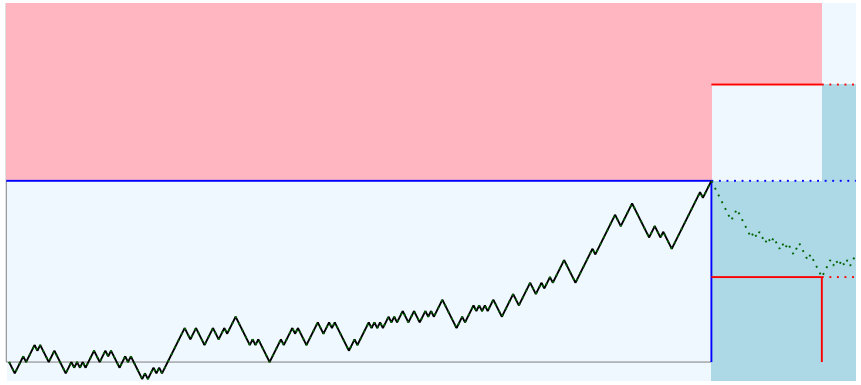

Figure 6: Result of the unfold based at $u<v$

Shaded regions: the unfolded path f_1 does not enter these regions.

4. Repeat this procedure of maximal unfolding to generate $f_2, f_3, \ldots$.
   Here are the accumulated results of five maximal unfolds.

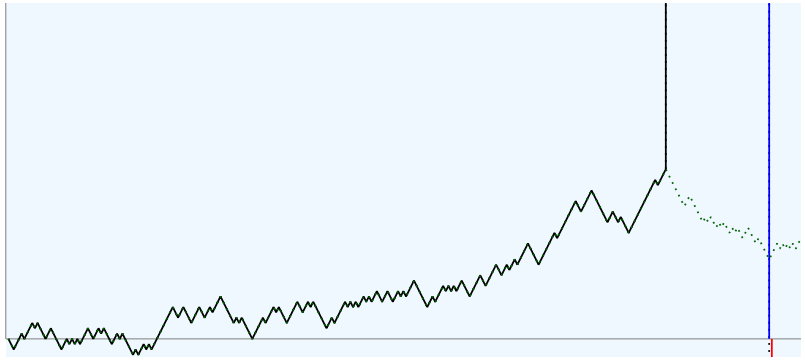

Figure 7: Second maximal unfold

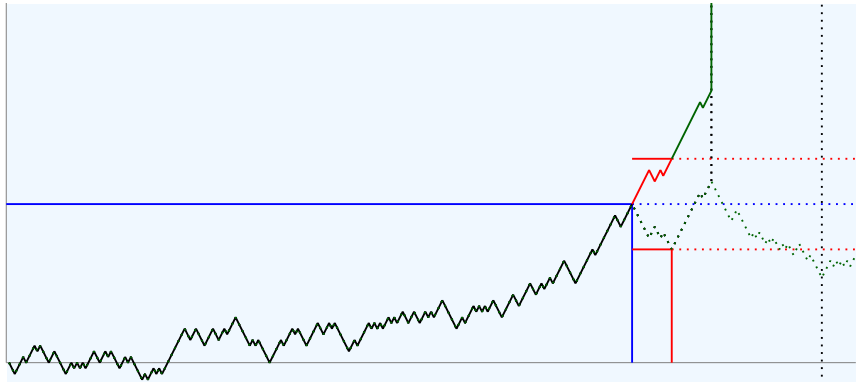

Figure 8: Third maximal unfold

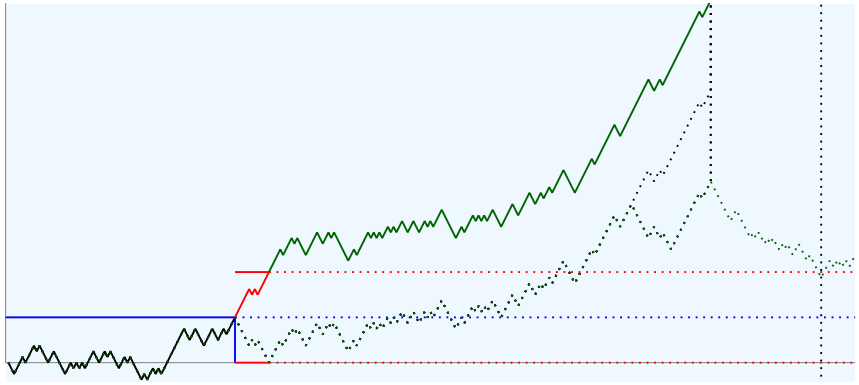

Figure 9: Fourth maximal unfold

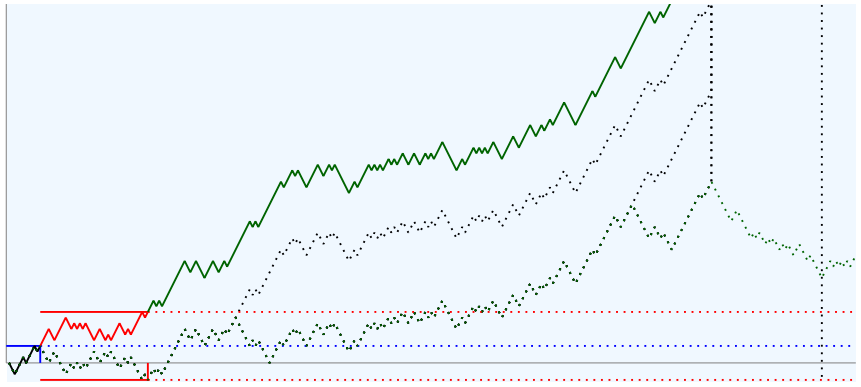

Figure 10: Fifth maximal unfold

Seeking a visual proof (considerations)

5. If $\sup_{u \leq v} (f_k(u) - f_k(v)) = 0$ for some finite k then the sequence ends at finite k.

- In this case f_k must be monotonic non-decreasing.
- In fact it must be monotonic **strictly** increasing (because the f_i all have $|\dot{f}_i| = 1$ almost everywhere and so can have no intervals of constancy):
- therefore $f_k(x) = x$ for all $x \in [0, L]$ and the Banach representation holds trivially for f_k.
- Invariance under vertical shifts and vertical reflections then implies the Banach representation then holds for the original f.
- (All this occurs precisely in the case of polygonal f!)

6. If the sequence continues indefinitely then the sequence of $f_n(x)$ is non-decreasing for all $x \in [0, L]$;
   - the interval of reflection (u, v) of the nth unfold must be "nested" either to the left, or to the right, or inside the interval of reflection of each previous n^{th} unfold;
   - the f_n become progressively more and more "monotonic increasing" with a monotonic non-decreasing limit;
   - an unfold based at $u < v$ destroys one local maximum (at u) and one local minimum (at v), and preserves all other local maxima and minima (at most countably many, since f is continuous);
   - the sequence of unfold sizes is non-increasing and summable;
   - after the infinite sequence of unfolds is completed, only degenerate local maxima and minima can survive for the limit f_∞ (others will be employed and thus destroyed at various stages of construction of the sequence);

7. **Final technical step:**
   - Exploit nesting to generate a martingale argument that $\dot{f}_n$ converges in L^1-norm to the constant function 1.
   - Thus $f_n(x) \to x$ as $n \to \infty$.
   - Re-write the roots x of $f_n(x) = y$ to give information about the point patterns determined as roots x of $f(x) = y$:

$$\mathbb{I}\{\text{ there is } x \in [0, L] \ : \ f_n(x) = y\}$$

can be unravelled to produce partial counts of the point patterns

$$\{x \in [0, L] \ : \ f(x) = y\} \ .$$

The counts increase monotonically to

$$\#\{x \in [0, L] \ : \ f(x) = y\} \ .$$

   - Ignore reflection zones of f_n to approximate Variation (f): Banach's representation follows by monotone convergence.

Conclusion

1. Ayari & Dubuc (1997) explain how to derive Crofton formula using Cauchy (1832) idea: project curve onto line of angle θ, use Banach representation to determine variation for projected curve, integrate out θ (appealing to definition of variation using polygonal approximations).
2. Unravelling of roots x of $f_n(x) = y$ needs concise rigorous expression;
3. Alternate visual proof of Banach representation uses "monotonic runs" (peeling off successive layers of graph of f as viewed from the left);
4. It would be interesting to get a *direct* visual proof of the general Crofton formula (and – even more interesting – of the Poincaré kinematic formula).

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Project details

`git` version and branch

Version (branch):	v0.0.0.5 (`main`)
Committed by:	Wilfrid Kendall <W.S.Kendall@warwick.ac.uk>
Date:	Mon Mar 16 13:56:16 2026 +0000

Images

Subject	Source	Licence
Georges-Louis Leclerc, Comte de Buffon (1707-1788)	François-Hubert Drouais - Musée Buffon à Montbard	Public Domain
Augustin-Louis Cauchy, (1789-1857)	Library of Congress.	PD-Art
Morgan William Crofton (1826–1915)	Unknown author, 1 January 1910	Public Domain
Sledgehammer	Slowking Man (also others) Own work	CC BY-SA 2.0

Previous versions of this talk

Date	Location	Occasion
13 March 2026	Warwick Statistics	Applied probability seminar

Summary of recent changes

Finalized talk version of 13 March 2026 at Warwick

To-do:

- ☒ Set up `git` repository;
- ☒ Finalize talk version;
- ☐ Display handout on web;
- ☐ Figures:
  - ☒ add details of images;
  - ☐ create direct from `quarto`;
  - ☐ no-go rectangles! (implement as pair meeting at each pivot)